Unit 2: Overview of Analog Circuits (Operational Amplifiers)

1. Operational Amplifiers (Op-Amp): Ideal vs Practical

> **Definition: An Operational Amplifier (Op-Amp) is a highly stable, high-gain, directly coupled DC voltage amplifier with a differential input and, usually, a single-ended output. It was originally designed to perform mathematical operations (addition, integration, etc.) in analog computers.**

An Op-Amp is the backbone of analog electronics. The most famous and widely used Op-Amp IC is the **IC 741**. It has two input terminals:

- **Inverting Input (-):** A signal applied here is amplified but inverted (180° phase shift) at the output.
- **Non-Inverting Input (+):** A signal applied here is amplified and remains in phase (0° phase shift) at the output.

It also requires dual power supplies ($+V_{cc}$ and $-V_{ee}$) to operate. The Op-Amp amplifies the *difference* between the two input voltages: $V_{out} = A (V_1 - V_2)$, where A is the open-loop gain.

Ideal Op-Amp vs. Practical Op-Amp Parameters:

To design circuits mathematically, we assume an "Ideal" Op-Amp. However, real-world (Practical) Op-Amps have limitations.

Parameter	Ideal Op-Amp	Practical Op-Amp (IC 741)
Voltage Gain (A_v)	Infinite (∞)	Very High (approx. 10^5 to 10^6)
Input Impedance (Z_{in})	Infinite (∞)	Very High ($2 M\Omega$)
Output Impedance (Z_{out})	Zero (0Ω)	Very Low (75Ω)
Bandwidth	Infinite (∞)	Limited (approx. 1 MHz)
CMRR (Common Mode Rejection Ratio)	Infinite (∞)	Very High (approx. 90 dB)
Slew Rate	Infinite (∞)	0.5 V/μs

Virtual Ground Concept:

Because the ideal Op-Amp has infinite gain, the voltage difference between the two input terminals must be zero ($V_1 - V_2 = 0 \Rightarrow V_1 = V_2$) for the output to be finite. If the non-inverting terminal is grounded ($0V$), the inverting terminal is also effectively at $0V$. This is called a **Virtual Ground**.

```
OP-AMP SYMBOL:
           +Vcc
             |
       (-) --|-\
 Inverting   |  \
             |   \_______ Output (Vo)
Non-Inv      |   /
       (+) --|+ /
             |
           -Vee
```

> 👉 **Hinglish Explanation:** Op-Amp (IC 741) ek bahut powerfull amplifier hai jiska use math ke calculations (plus, minus, integration) karne ke liye hota tha. Ideal op-amp ek sapna hai jo 100% perfect hota hai (Gain infinite, Input resistance infinite). Par real life me (Practical op-amp) inki values infinite na hokar sirf 'bahut zyada' hoti hain. Virtual ground ka matlab hai ki op-amp ke dono input pins ka voltage hamesha ek-dusre ke barabar hota hai.

2. Open Loop and Closed Loop Configurations

Definition: An Op-Amp can be connected in a circuit with or without feedback. "Open Loop" means there is no connection between the output and input. "Closed Loop" means a portion of the output signal is fed back to the input.

1. Open Loop Configuration:

In this configuration, no feedback resistor is connected from the output to the input pins. The Op-Amp operates at its maximum natural gain (Open-loop gain, $A_{ol} \approx 10^5$).

- **Behavior:** Because the gain is so massive, even a tiny microvolt difference between the inputs will be multiplied by 10^5, pushing the output voltage to its maximum limit instantly ($+V_{sat}$ or $-V_{sat}$).
- **Applications:** Used mainly as a **Comparator** (to check which of the two input voltages is higher) and in zero-crossing detectors. It is *never* used as a linear amplifier because it becomes highly unstable.

2. Closed Loop Configuration:

Here, a feedback path is provided from the output terminal back to the input terminal using a resistor (R_f).

- **Negative Feedback:** If the output is connected back to the *Inverting (-)* terminal, it is negative feedback. This opposes the input signal, reducing the overall gain.
- **Advantages of Negative Feedback:**
  1. Controls and stabilizes the voltage gain to an exact desired value.
  2. Increases Bandwidth (frequency response).
  3. Increases input impedance and decreases output impedance.
  4. Reduces noise and distortion.
- **Applications:** Used in amplifiers, filters, adders, and integrators. (Positive feedback is used only in Oscillators).

```
OPEN LOOP (No Feedback)         CLOSED LOOP (Negative Feedback)
```

```
     ___                          ___
 V1 -|-\                      V1 -|-\
     |  \                          |  \
 V2 -|+  \_____ Vo            V2 -|+  \_____ Vo
     |___/                          |___/   |
                                     |      |
                                     +--[Rf]+
```

👉 **Hinglish Explanation:** Open loop ka matlab hai bina kisi lagam ke ghoda. Isme output se input tak koi rasta nahi hota. Gain itna high hota hai ki thodi si input dene par bhi output full limit (saturate) par pahunch jata hai, isliye isko as a "Comparator" use karte hain. Closed loop me hum output ka thoda hissa resistor lagakar wapas inverting pin (-) me bhej dete hain (Negative feedback). Isse gain control me aa jata hai aur circuit stable ho jata hai.

3. Op-Amp as an Amplifier (Inverting & Non-Inverting)

Definition: By using negative feedback, an Op-Amp can be configured as a linear voltage amplifier. It scales up the input voltage by a fixed mathematical ratio. It has two main configurations: Inverting and Non-Inverting.

1. Inverting Amplifier:

The input signal (V_{in}) is applied to the Inverting (-) terminal through an input resistor (R_{in}), and the Non-Inverting (+) terminal is grounded. A feedback resistor (R_f) connects the output to the inverting input.

- **Phase Shift:** The output signal is exactly 180° out of phase with the input signal (it is flipped upside down).
- **Voltage Gain (A_v):** $A_v = - \frac{R_f}{R_{in}}$

Output Voltage: $V_{out} = - \left(\frac{R_f}{R_{in}} \right) V_{in}$

(The negative sign indicates the 180-degree phase shift).

2. Non-Inverting Amplifier:

The input signal (V_{in}) is applied directly to the Non-Inverting (+) terminal. The Inverting (-) terminal is connected to ground through R_{in}, and also receives feedback from the output via R_f.

- **Phase Shift:** The output signal is strictly in phase with the input signal (0° phase shift).
- **Voltage Gain (A_v):** $A_v = 1 + \frac{R_f}{R_{in}}$

Output Voltage: $V_{out} = \left(1 + \frac{R_f}{R_{in}} \right) V_{in}$

Note: The gain of a non-inverting amplifier can never be less than 1.

```
INVERTING AMPLIFIER:
         Rf
      +--\/\/--+
      |        |
```

```
Vin --\/\/--|-\|
      Rin   |  \
            |   \____ Vout
GND --------|+  /
            |__/

NON-INVERTING AMPLIFIER:
         Rf
      +--\/\/--+
      |        |
GND --\/\/--|-\|
      Rin   |  \
            |   \____ Vout
Vin --------|+  /
            |__/
```

👉 **Hinglish Explanation:** Amplifier ka kaam signal ko bada karna hai. **Inverting** wale me input (-) pin par dete hain, toh output bada hokar ulta (reverse) ho jata hai. Iska formula $-R_f/R_{in}$ hota hai. **Non-Inverting** me input (+) pin par dete hain, toh output bada hota hai par seedha hi rehta hai. Iska formula $1 + (R_f/R_{in})$ hota hai. Formulas numericals ke liye bahut important hain!

4. Application of Op-Amp as an Adder (Summing Amplifier)

Definition: A Summing Amplifier (or Adder) is a closed-loop Op-Amp circuit that produces an output voltage which is proportional to the algebraic sum of multiple input voltages.

The adder circuit is essentially an extension of the Inverting Amplifier. Instead of a single input resistor, multiple input resistors are connected in parallel to the inverting (-) terminal. The non-inverting (+) terminal is grounded.
Let three input voltages $V_1, V_2,$ and V_3 be applied through input resistors $R_1, R_2,$ and R_3 to the inverting terminal. A feedback resistor R_f connects the output to the inverting terminal.
By applying Kirchhoff's Current Law (KCL) at the virtual ground node:

$$V_{out} = - \left(\frac{R_f}{R_1} V_1 + \frac{R_f}{R_2} V_2 + \frac{R_f}{R_3} V_3 \right)$$

Special Case (Equal Resistors):

If all the resistors are made equal, i.e., $R_1 = R_2 = R_3 = R_f = R$, the formula simplifies significantly:

$$V_{out} = -(V_1 + V_2 + V_3)$$

Here, the Op-Amp acts as a pure mathematical adder (with an inverted sign).

Applications:

- **Audio Mixers:** Combining signals from multiple microphones or instruments into a single audio track.
- **Digital-to-Analog Converters (DAC):** Summing binary inputs weighted by specific resistor values to create an analog output.

```
SUMMING AMPLIFIER (ADDER):

           Rf
      +---\/\/---+
      |          |
V1 --\/\/--+     |
     R1    |     |
V2 --\/\/--+--|-\|
     R2    |  |  \
V3 --\/\/--+  |   \____ Vout
     R3       |   /
GND ----------|+ /
              | /
              |/
```

Hinglish Explanation: Adder circuit ka naam hi sab batata hai—ye alag-alag voltages ko add (plus) karta hai. Ye inverting amplifier ka hi bada bhai hai. Isme minus (-) pin par ek se zyada voltages jod dete hain. Jaise music studio me 3 alag-alag mikes ki awaaz ek sath mix karke ek speaker se nikalni ho, toh Summing amplifier ka use hota hai. Agar sab resistors ki value same rakh de toh output teeno voltages ka direct sum (minus sign ke sath) aayega.

5. Application of Op-Amp as a Differentiator

Definition: A differentiator is an Op-Amp circuit whose output voltage is directly proportional to the rate of change (time derivative) of the input voltage.

The circuit is constructed by modifying an inverting amplifier: the input resistor (R_{in}) is replaced by a **Capacitor (C)**, while the feedback resistor (R_f) remains in place.
Working Principle: Because the current flowing through a capacitor is proportional to the rate of change of voltage across it ($I_c = C \frac{dV}{dt}$), and this same current flows through the feedback resistor R_f, the output voltage becomes proportional to the derivative of the input.

$$V_{out} = - R_f C \frac{d(V_{in})}{dt}$$

Waveform Shaping (Crucial for Exams):

A differentiator modifies the shape of alternating input signals drastically:

- If Input is a **Square Wave** $\rightarrow$ Output is a series of **Sharp Spikes (Needles)**.
- If Input is a **Triangular Wave** $\rightarrow$ Output is a **Square/Rectangular Wave**.
- If Input is a **Sine Wave** $\rightarrow$ Output is a **Cosine Wave** (shifted by 90°).

Applications:

- **Wave Shaping Circuits:** Used to generate sharp trigger pulses from square waves for timing circuits.
- **High-Pass Filters:** Acts as an active high-pass filter because capacitive reactance decreases at high frequencies.
- **PID Controllers:** Provides the "Derivative" action in industrial control systems to predict errors.

```
DIFFERENTIATOR CIRCUIT:

           Rf
      +---\/\/---+
      |          |
Vin --| |---+--|-\
      C     |  |  \
            |  |   \____ Vout
GND --------+--|+  /
               |  /
               | /

Input: Square Wave  Output: Spikes
  __      __          |        |
 |  |    |  |        -|-      -|-
_|  |____|  |   -->   |        |
                       |        |
```

👉 **Hinglish Explanation:** Differentiator math ka 'Differentiation' (derivative) karta hai. Iska circuit yaad rakhne ki trick: **C** pehle, **R** baad me (feedback me). Ye signal ki speed (rate of change) check karta hai. Agar hum isme square wave denge, jahan achanak change hota hai, ye wahan ek lambi sui (sharp spike) paida karega. Ye circuit achanak aane wale changes ko pakadne (trigger pulse banane) ke kaam aata hai.

6. Application of Op-Amp as an Integrator

Definition: An integrator is an Op-Amp circuit whose output voltage is directly proportional to the time integral (area under the curve) of the input voltage.

The circuit is constructed by modifying an inverting amplifier: the input is a **Resistor (R)**, but the feedback component is replaced by a **Capacitor (C)**. It is the exact structural opposite of the Differentiator.
Working Principle: The input current $I_{in} = V_{in}/R$ flows entirely into the feedback capacitor C, charging it. The voltage across a capacitor is proportional to the integral of current over time ($V_c = \frac{1}{C} \int I dt$). Therefore, the output voltage represents the integral of the input.

$$V_{out} = - \frac{1}{R C} \int_{0}^{t} V_{in} dt + V_{initial}$$

Waveform Shaping (Crucial for Exams):

An integrator also modifies the shape of alternating input signals:

- If Input is a constant **DC Voltage** $\rightarrow$ Output is a **Ramp (linear slope)**.
- If Input is a **Square Wave** $\rightarrow$ Output is a **Triangular Wave**.
- If Input is a **Sine Wave** $\rightarrow$ Output is a **Negative Cosine Wave** (shifted by 90°).

Applications:

- **Analog Computers:** Historically used to solve complex differential equations in real-time.
- **Analog-to-Digital Converters (ADCs):** Specifically in dual-slope integrating ADCs used in digital multimeters.

- **Low-Pass Filters:** Acts as a low-pass filter because capacitive reactance increases at low frequencies.
- **Function Generators:** Used to generate perfectly linear triangular waves from square wave oscillators.

```
INTEGRATOR CIRCUIT:

           C
      +---||-----+
      |          |
Vin --\/\/--+--|-\
      R     |  |  \
            |  |   \____ Vout
GND --------+--|+  /
               |  /
               | /

Input: Square Wave  Output: Triangular
  __      __           /\      /\
 |  |    |  |         /  \    /  \
_|  |____|  |   --> _/    \__/    \_
```

Hinglish Explanation: Integrator math ka 'Integration' karta hai. Differentiator ka bilkul ulta: isme **R** pehle aur **C** baad me (feedback me) lagta hai. Ye input ke area ko add (accumulate) karta rehta hai. Agar aap ek square wave denge, toh ye dheere-dheere charge aur discharge hoga, jisse output me ekdam perfect **Triangular wave** milegi. Ye analog computers aur digital multimeters (ADC) me bahut use hota hai.

Quick Revision Notes (Exam Saver)

- **Ideal Op-Amp:** Infinite Gain, Infinite Input Resistance, Zero Output Resistance, Infinite Bandwidth.
- **Virtual Ground:** The potential at the inverting (-) pin tracks the non-inverting (+) pin due to infinite gain.
- **Inverting Amplifier:** $A_v = -R_f/R_{in}$. Phase shift = 180°.
- **Non-Inverting Amplifier:** $A_v = 1 + (R_f/R_{in})$. Phase shift = 0°.
- **Adder:** Adds multiple voltages connected to the (-) terminal. $V_{out} = -(V_1 + V_2 + V_3)$.
- **Differentiator:** Capacitor at input, Resistor in feedback. Square wave in $\rightarrow$ Spike out. Rate of change.
- **Integrator:** Resistor at input, Capacitor in feedback. Square wave in $\rightarrow$ Triangle out. Accumulates area.

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